Math Inquirers

Extension Workbook

with answers

Level 4

Self-study | Ages 9-11



My name _____

About this book

Math Inquirers Extension Workbook, Level 4
Self-study practice with answers.

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- This is a private-enrichment book. It is not an official school or examination book, and it makes no promise about any test result.
- We have checked the answers with care. If you find a mistake, please tell us so we can fix it.
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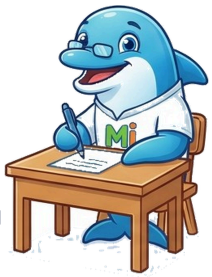
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Welcome!



This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 15: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- Printing this soft copy? Print at 100 percent so the rulers are true size.

The four page names

- ● **Start here.** The main path. Do these first.
- ◆ **Think harder.** Longer problems. Hints are printed upside down.
- ★ **Go further.** A weekend page. Big challenges, just for fun.
- ✓ **Check yourself.** Four quick questions at the end.

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2	**Big number puzzles** Not in this proof edition	–
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My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

Week	Unit	Session 1 Learn it, Start here	Session 2 Start here	Session 3 Think harder, Check yourself	Weekend page Go further	Done
1	Factor hunters	Not in this proof edition				☐
2	Big number puzzles	Not in this proof edition				☐
3	Operation detectives	Not in this proof edition				☐
4	Fraction frontier	Not in this proof edition				☐
5	Decimal Detectives	6-7	8-9	10 and 12	11	☐
6	Angle turns	Not in this proof edition				☐
7	Area architects II	Not in this proof edition				☐
8	Symmetry and nets	Not in this proof edition				☐
9	Data detectives	Not in this proof edition				☐
10	Puzzle day	Not in this proof edition				☐
11	Catch-up week	Repeat any Check yourself where your score was 2 or less.				☐

Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

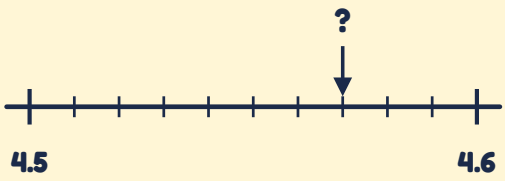
Learn it: Decimals

My goal. I can read decimals on number lines, compare and order them and round them. I can solve money and mass puzzles with bar models.

On a number line, first find what one small step is worth. To compare decimals, line up the decimal points and compare place by place from the left.

Worked example 1

What number does the arrow show?



1. 4.5 to 4.6 has 10 steps. 1 step = 0.01.
2. The arrow is 7 steps after 4.5.

Answer: The arrow shows 4.57.

Worked example 2

A number has 2 decimal places. It rounds to 6.3 to 1 decimal place. What are the least and greatest it can be?



1. 6.25 is halfway, so it rounds up to 6.3.
2. 6.35 rounds up to 6.4, so the last number is 6.34.

Answer: Least 6.25 and greatest 6.34.

How to do it

1. Number line: work out what 1 small step is worth.
2. Compare: line up the points. Add zeros to fill the gaps.
3. Round: find the 2 neighbours. Halfway rounds up.

Place value

ones	.	tenths	hundredths	thousandths
3	.	4	0	5

- 3.405 has 4 tenths, 0 hundredths and 5 thousandths.

Your turn

Answer at the back of the book.

Try Round 8.46 to 1 decimal place. ☐

Start here

Do these first. Take your time.

1 **Each small step on the number line is the same size. Write the number each arrow shows.**



(a) A = ☐ (b) B = ☐

2 **Now each small step is 0.01. Write the number each arrow shows.**



(a) C = ☐ (b) D = ☐

(c) Which arrow shows a number that rounds to 0.5? C D

3 **Each small step is 0.001. Write the number the arrow shows.**



(a) E = ☐

Start here

Keep going. Check your work.

4 **Circle the correct sign.**

(a) 4.06 > < = 4.6

(b) 0.38 > < = 0.308

(c) 12.5 > < = 12.50

(d) 7.09 > < = 7.1

5 **Write the numbers in order. Begin with the smallest.**

5.205 5.25 5.052 5.5 5.025

_____

6 **Write the value of the digit 6 in each number.**

(a) 4.562: ☐ (b) 6.043: ☐ (c) 0.936: ☐

7 **Round 6.847.**

(a) To the nearest whole number: ☐

(b) To 1 decimal place: ☐

(c) To 2 decimal places: ☐

Working: _____

8 **Ali buys a ruler for $1.85 and a notebook for $2.40. He pays with a $5 note. How much change does he get? $** ☐

Working: _____

_____

Start here

Nearly there. You can do it.

9 **Ravi and Kumar have $12.60 altogether. Ravi has $2.40 more than Kumar.**



(a) Kumar has $ ☐ (b) Ravi has $ ☐

10 **A box holds 4 packs of rice. All the packs have the same mass. The box and packs together have a mass of 4.1 kg. The empty box has a mass of 0.5 kg.**

(a) The 4 packs together: ☐ kg

(b) One pack: ☐ kg

(c) Kumar takes out 1 pack. The box and packs now have a mass of ☐ kg

Working: _____

11 **Work these out.**

(a) 4.7 + 2.35 = ☐

(b) 9.2 – 3.64 = ☐

(c) 2.5 × 4 = ☐

(d) 7.2 ÷ 6 = ☐

Working: _____

12 **Write each as a decimal.**

(a) 7/10 = ☐ (b) 19/100 = ☐ (c) 3 1/2 = ☐

(d) 2 3/4 = ☐

Think harder

Hints are upside down. Try first!

1 **A number has 3 decimal places. Rounded to 2 decimal places it is 4.27.**

(a) The least it can be: ☐

(b) The greatest it can be: ☐

Working: _____

Hint. Find the halfway marks next to 4.27. Which ones round to 4.27?

2 **Use the digit cards 1, 4, 6 and 9 and a decimal point. Use every card once. The point goes between two digits. Make the number that is closest to 20.**

☐

List: _____

Hint. List the numbers that start with 1 and with 9. Which are near 20?

3 **A mug costs $2.40 more than a cup. 3 mugs and 2 cups cost $17.20 altogether.**

(a) A cup costs $ ☐

(b) A mug costs $ ☐

Draw a bar model here.

Hint. Change the 3 mugs into cups. How much extra is that?

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **A full bottle of juice has a mass of 1.26 kg. When one third of the juice is left, the bottle has a mass of 0.58 kg. What is the mass of the empty bottle?**

☐ kg

Working: _____

_____

Hint. The juice that was drunk is 2/3 of the juice. Its mass is 1.26 − 0.58.

2 **Mei has 12 coins. They are all 10¢ coins and 50¢ coins. Altogether they are worth $3.60. How many 50¢ coins does she have?** ☐

Working: _____

_____

Hint. Suppose all 12 coins are 10¢. How much money is that?

3 **Four children share $20.40. Each child gets $0.60 more than the one before. The youngest gets the least.**

(a) The youngest gets $ ☐

(b) The oldest gets $ ☐

Working: _____

Hint. Take away the extras first: 0.60, 1.20 and 1.80.

4 **Beyond Level 4: Number patterns with decimals**
0.35, 0.40, 0.45, 0.50, and so on.

(a) The 20th number is ☐

(b) Which place in the pattern is 2.05? Place ☐

Working: _____

Hint. Each place adds 0.05. How many steps after the 1st?

✔ Check yourself

No hints. Try each one on your own.

1 **Write the numbers in order. Begin with the smallest.**
6.07 6.7 6.007 6.17

_____

Working: _____

2 **Round 7.846 to the nearest whole number ☐, 1 decimal place ☐, 2 decimal places ☐**

Working: _____

3 **Siti buys 4 notebooks at $1.35 each and pays $10. Her change is $ ☐**

Working: _____

4 **A number with 2 decimal places rounds to 3.6 to 1 decimal place. The least it can be is ☐ and the greatest is ☐.**

Working: _____

My score: ☐ out of 4. Check the answers at the back.

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

My progress

Tick each box when you finish. Write your Check yourself score.

Unit	Topic	Learn it	Start here	Think harder	Go further	Score	Date
1	Factor hunters	☐	☐	☐	☐	/ 4	
2	Big number puzzles	☐	☐	☐	☐	/ 4	
3	Operation detectives	☐	☐	☐	☐	/ 4	
4	Fraction frontier	☐	☐	☐	☐	/ 4	
5	Decimal Detectives	☐	☐	☐	☐	/ 4	
6	Angle turns	☐	☐	☐	☐	/ 4	
7	Area architects II	☐	☐	☐	☐	/ 4	
8	Symmetry and nets	☐	☐	☐	☐	/ 4	
9	Data detectives	☐	☐	☐	☐	/ 4	
10	Puzzle day	☐	☐	☐	☐	/ 4	

Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____



Certificate of Completion

Well done,

You finished the Math Inquirers Extension Workbook, Level 4.

Date

Signed by a grown-up

Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

Unit 5 · Decimal Detectives

Your turn
page 6

Answer: 8.5
Why: 8.46 is between 8.4 and 8.5. The halfway mark is 8.45. 8.46 is after it, so it rounds up to 8.5.

START HERE

1
page 7

Answer: (a) 5.3 (b) 5.8
Why: From 5 to 6 there are 10 steps, so each step is 0.1. A is 3 steps after 5: 5.3. B is 8 steps after 5: 5.8.

2
page 7

Answer: (a) 0.43 (b) 0.48 (c) D
Why: From 0.4 to 0.5 there are 10 steps, so each step is 0.01. C is 3 steps after 0.4: 0.43. D is 8 steps after 0.4: 0.48. 0.48 is past the halfway mark 0.45, so it rounds to 0.5.

3
page 7

Answer: (a) 3.427
Why: From 3.42 to 3.43 there are 10 steps, so each step is 0.001. E is 7 steps after 3.42: 3.427.

4
page 8

Answer: (a) < (b) > (c) = (d) <
Why: (a) 4.06 has 0 tenths and 4.6 has 6 tenths. (b) 0.38 has 8 hundredths and 0.308 has 0 hundredths. (c) 12.50 is 12.5 with a 0 added. (d) 7.09 is 7.09 and 7.1 is 7.10, and 10 hundredths is more than 9.

5
page 8

Answer: 5.025, 5.052, 5.205, 5.25, 5.5
Why: Write each with 3 decimal places: 5.205, 5.250, 5.052, 5.500, 5.025. All have 5 ones. Compare the tenths: 5.025 and 5.052 have 0 tenths, then 5.205 and 5.250 have 2 tenths, then 5.500. 5.025 comes before 5.052, and 5.205 comes before 5.250.

6
page 8

Answer: (a) 0.06 (b) 6 (c) 0.006
Why: In 4.562 the 6 is in the hundredths place: 0.06. In 6.043 it is in the ones place: 6. In 0.936 it is in the thousandths place: 0.006.

7
page 8

Answer: (a) 7 (b) 6.8 (c) 6.85
Why: (a) 6.847 is between 6 and 7 and is past 6.5, so 7. (b) 6.847 is between 6.8 and 6.9. The halfway mark is 6.85, so it rounds to 6.8. (c) 6.847 is between 6.84 and 6.85. The halfway mark is 6.845, so it rounds up to 6.85.

8
page 8

Answer: 0.75
Why: $1.85 + $2.40 = $4.25. $5.00 − $4.25 = $0.75, which is 75¢.

9
page 9

Answer: (a) 5.10 (b) 7.50
Why: Take away the extra $2.40: $12.60 − $2.40 = $10.20. Two equal parts are $10.20, so Kumar has $10.20 ÷ 2 = $5.10. Ravi has $5.10 + $2.40 = $7.50.

10
page 9

Answer: (a) 3.6 (b) 0.9 (c) 3.2
Why: 4.1 − 0.5 = 3.6 kg for the 4 packs. 3.6 ÷ 4 = 0.9 kg for 1 pack. After 1 pack is taken out, the mass is 4.1 − 0.9 = 3.2 kg.

11 page 9
Answer: (a) 7.05 (b) 5.56 (c) 10 (d) 1.2
Why: (a) 4.70 + 2.35 = 7.05. (b) 9.20 − 3.64 = 5.56. (c) 2.5 × 4 = 10. (d) 7.2 ÷ 6 = 1.2, because 6 × 1.2 = 7.2.

12 page 9
Answer: (a) 0.7 (b) 0.19 (c) 3.5 (d) 2.75
Why: (a) 7 tenths is 0.7. (b) 19 hundredths is 0.19. (c) 1/2 = 5/10 = 0.5, so 3.5. (d) 3/4 = 75/100 = 0.75, so 2.75.

THINK HARDER

1 page 10
Answer: (a) 4.265 (b) 4.274
Why: 4.265 is halfway between 4.26 and 4.27, so it rounds up to 4.27. 4.275 would round up to 4.28, so the greatest is 4.274.

2 page 10
Answer: 19.64
Why: Numbers just below 20 start with 19: 19.64 and 19.46. 19.64 is 0.36 below 20 and 19.46 is 0.54 below. Numbers that start with 14 or 16 are further away, and the numbers above 20 start with 41, 46, 49, 61 and so on, which are much further.

3 page 10
Answer: (a) 2.00 (b) 4.40
Why: If every mug cost the same as a cup, the 5 items would cost $17.20 − 3 × $2.40 = $17.20 − $7.20 = $10.00. So a cup costs $10.00 ÷ 5 = $2.00 and a mug costs $4.40. Check: 3 × $4.40 + 2 × $2.00 = $13.20 + $4.00 = $17.20.

GO FURTHER

1 page 11
Answer: 0.24
Why: The juice that was drunk has a mass of 1.26 − 0.58 = 0.68 kg, and that is 2 thirds of the juice. 1 third is 0.34 kg, so all the juice is 1.02 kg. The empty bottle is 1.26 − 1.02 = 0.24 kg.

2 page 11
Answer: 6
Why: If all 12 coins were 10¢, the total would be $1.20. The real total is $2.40 more. Changing a 10¢ coin into a 50¢ coin adds 40¢. $2.40 ÷ 40¢ = 6 coins. So she has 6 coins of 50¢ and 6 coins of 10¢. Check: $3.00 + $0.60 = $3.60.

3 page 11
Answer: (a) 4.20 (b) 6.00
Why: The extras are $0 + $0.60 + $1.20 + $1.80 = $3.60. Without them the 4 children share $20.40 − $3.60 = $16.80, so the youngest gets $16.80 ÷ 4 = $4.20. The oldest gets $4.20 + $1.80 = $6.00. The amounts are $4.20, $4.80, $5.40 and $6.00.

4 page 11
Answer: (a) 1.3 (b) 35
Why: The pattern adds 0.05 each time. The 20th number is 0.35 + 19 × 0.05 = 0.35 + 0.95 = 1.30. For 2.05, the number of steps from 0.35 is (2.05 − 0.35) ÷ 0.05 = 34, so it is the 35th number.

CHECK YOURSELF

1
page 12

Answer: 6.007, 6.07, 6.17, 6.7
Why: With 3 decimal places: 6.070, 6.700, 6.007, 6.170. In order: 6.007, 6.070, 6.170, 6.700.

2
page 12

Answer: 8, 7.8, 7.85
Why: 7.846 is past 7.5, so 8. It is below 7.85, so 7.8. It is past 7.845, so 7.85.

3
page 12

Answer: 4.60
Why: 4 × $1.35 = $5.40. $10.00 – $5.40 = $4.60.

4
page 12

Answer: Least 3.55, greatest 3.64
Why: 3.55 is halfway between 3.5 and 3.6, so it rounds up to 3.6. 3.65 would round up to 3.7, so the greatest is 3.64.